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①

+ s^6	1	-9	-28	96
- s^5	-2	6	56	
+ s^4	x=6	y=0	96	
+ s^3	z=6	t=88		
- s^2	y=-88	96		
+ s^1	$\Delta=\frac{88}{11}$			
+ s	96			

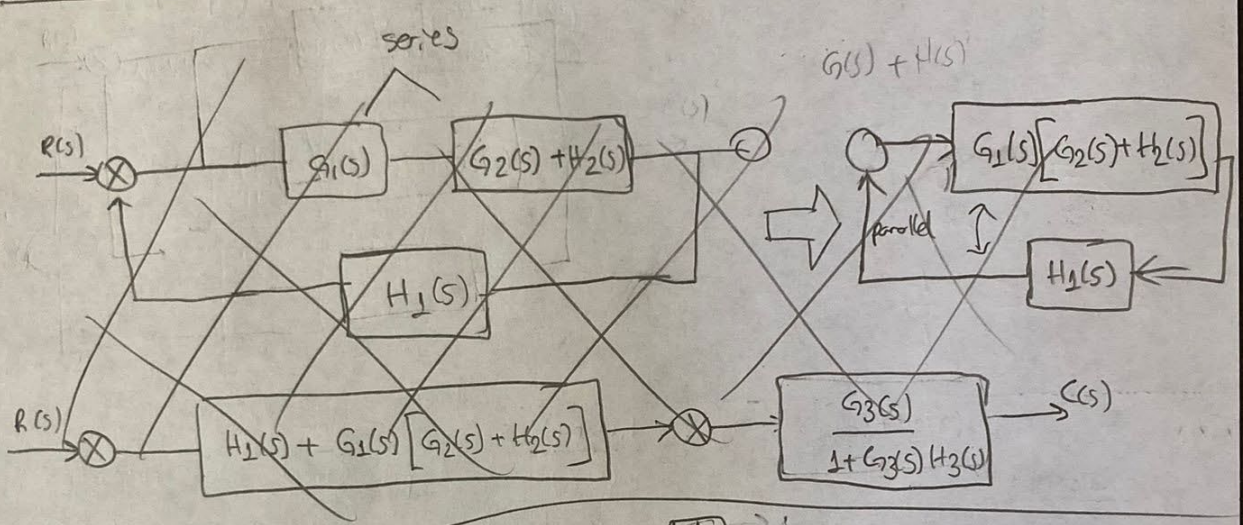
$x=\frac{18-6}{-2}=6 \quad y=\frac{56-56}{-2}=0 \quad z=\frac{36-0}{6}=6 \quad t=\frac{336+192}{6}=88$

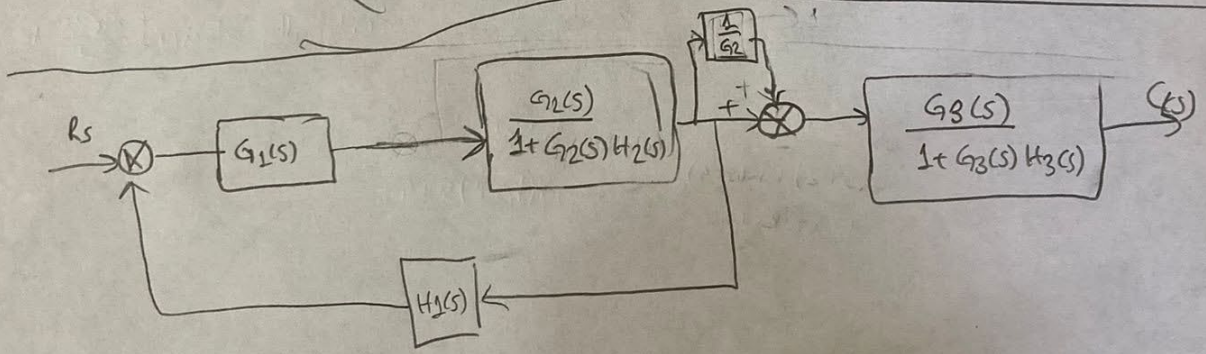
$y=\frac{0-88.6}{6}=-88 \quad \Delta=\frac{(-88.88+56.6}{-88}=88+\frac{72}{11}$

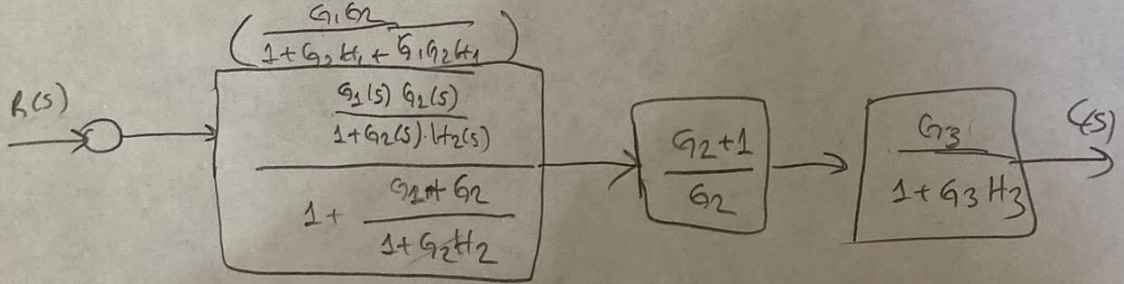
4 sign changes 4 poles in RHP

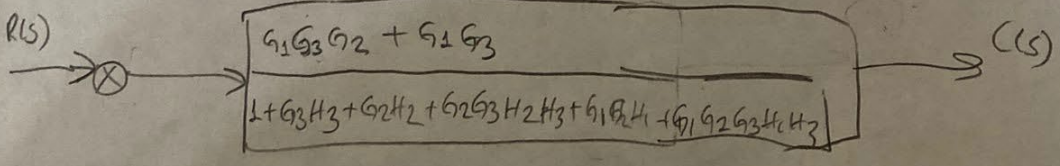
2 poles in LHP

③

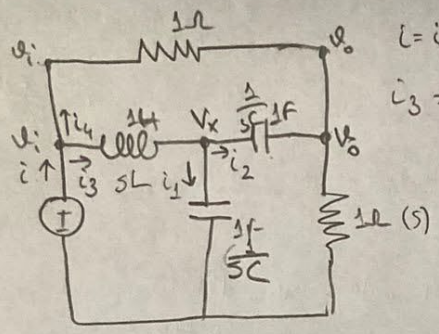








(2)



$i = i_3 + i_4$

$i_3 = i_1 + i_2$

$i_1 = C \cdot \frac{dV_x}{dt} = \frac{1}{sC} \cdot \frac{dV_x}{dt}$

$i_2 = \frac{V_o - V_x}{\frac{1}{sC}}$

$i_3 = \frac{V_x - V_i}{sL}$

$\frac{V_x - V_i}{sL} = \frac{dV_x}{sC.dt} + sC(V_o - V_x)$

$\rightarrow s$

$\| \rightarrow \frac{1}{s}$

$\rightarrow s$

$\underbrace{\quad}_{\text{output}} = \underbrace{\quad}_{\text{input}}$

$T.F = \frac{\text{output}}{\text{input}} = \frac{V_o}{V_i}$

---

(4) unity feed back $H(s)=1$ $\quad G(s) = \frac{K(s+4)}{s(s+2)}$

$T(s) = \frac{G(s)}{1 + G(s)H(s)}$

$T(s) = \frac{\frac{K(s+4)}{s(s+2)}}{1 + \frac{K(s+4)}{s(s+2)}} = \frac{K(s+4)}{s(s+2) + K(s+4)} = \frac{K(s+4)}{s^2 + (K+2)s + 4}$

$s^2 + 2s + Ks + 4$

Unit step response

critically damped $\zeta = 1$

$T(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$

$\omega_n^2 = 4$

$\omega_n = 2$

$\zeta = 1$

$2\zeta\omega_n = K+2$

$2.1.2 = K+2$

$4 = K+2$

$K = 2$

$T(s) = \frac{2s+8}{s^2 + 4s + 4}$ → zero $= -4$

→ pole $= -2, -2$

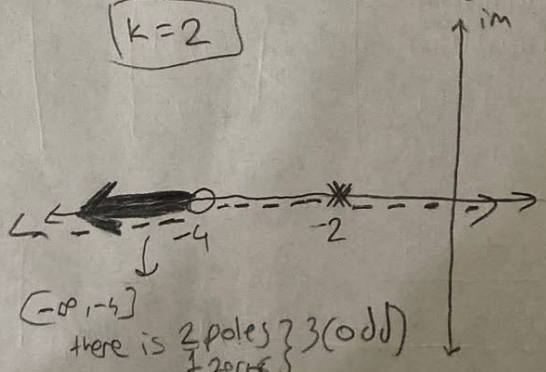
$(s+2)^2$

$\max(2,1) = 2$

2 branches

$\frac{180}{1} = 180$

the asymtote is horizontal



$(-\infty, -4]$

there is 2 poles } 3 (odd)
1 zero }

RLC

$$\ddot{U}_c(t) + \frac{R}{L}\dot{U}_c(t) + \frac{1}{LC}U_c(t) = \frac{1}{LC}U(t)$$

output / input

LAPLACE

$\delta \to 1$	$\sin\omega t = \frac{\omega}{s^2+\omega^2}$	$\frac{dy(t)}{dt} \to sF(s) - f(0)$	initial value of $f(t)$
$3 \to \frac{3}{s}$	$\cos\omega t = \frac{s}{s^2+\omega^2}$	$\int_0^t f(\tau)d\tau \to \frac{F(s)}{s}$	
$t \to \frac{1}{s^2}$	$e^{-at}\sin\omega t = \frac{\omega}{(s+a)^2+\omega^2}$	$\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s)$	
$e^{-qt} = \frac{1}{s+q}$	$e^{-at}\cos\omega t = \frac{s+a}{(s+a)^2+\omega^2}$	$e^{at}f(t) \to F(s-a)$	
		$f(t-\tau) \to e^{-\tau s}F(s)$	

$$\ddot{y} + 4\dot{y} + 5y = 0$$

↓ Laplace

$$s^2Y(s) - sy(0) - \dot{y}(0) + 4sY(s) - 4y(0) + 5Y(s) = 0$$

Transfer function

all initial conditions zero!

C.L.T.F.:

$$T.F. = \frac{G(s)}{1+G(s)H(s)}$$

$$G(s) = \frac{Y(s)}{U(s)} = \frac{K}{\tau s+1}$$

K: DC gain (s.state size)

τ: Time constant (quick)

step response

$$Y(s) = G(s)U(s)$$

$$y(s) = \frac{K}{\tau s+1}\cdot\frac{1}{s} \xrightarrow{\text{inverse}} y(t) = K\left(1-e^{-t/\tau}\right)$$

Second order Transfer Function

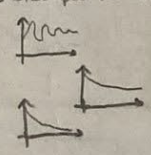
$$G(s) = \frac{Y(s)}{U(s)} = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$$

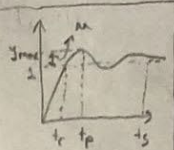
ζ → damping ratio ω_n → undamped natural freq.

$\zeta < 1$ → underdamped

$\zeta > 1$ → overdamped

$\zeta = 1$ → critically damped





$M = y_{max} - y_{ss}$

t_r → rise

t_p → peak

t_s → setting

$\zeta < 1$: $s_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$

$\zeta > 1$: $s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2-1}$

$\zeta = 1$: $s_{1,2} = -\omega_n$

$\zeta = 0$: $s_{1,2} = \pm j\omega_n$ (undamped)

$\zeta < 0$ (negatively damped)

Routhe tabulation

$$D(s) = s^4 + 2s^3 + 3s^2 + 4s + 5 = 0$$

s^4	1	3	5
s^3	2	4	
s^2	x	y	
s^1	z		
s^0	y		

$$x = \frac{1}{2}\begin{vmatrix}3 & 1\\ 4 & 2\end{vmatrix} \qquad y = \frac{1}{2}\begin{vmatrix}5\\ 3\end{vmatrix}$$

Sign change

There is 2 sign change → 2 poles in RHP

Total root 4 → 2 RHP, 2 LHP (unstable)

$$e_{ss} = \lim_{s\to 0} s\frac{1}{1+\frac{K}{s^N}}\cdot R(s) \qquad N = 1,2,3$$

step → ramp → parab.

N	K_p	K_v	K_a	r(t)= unit $R_0 1$	ramp $V_0 t$	parab $A_0 t^2/2$
0	K_p	0	0	$\frac{R_0}{1+K_p}$	∞	∞
1	∞	K_v	0	0	$\frac{V_0}{K_v}$	∞
2	∞	∞	K_a	0	0	$\frac{A_0}{K_a}$

(S. state error)

Properties

$$K = \frac{s-p_i}{s-z_i}$$

① $K=0 \to s=p_i$

$K=\infty \to s=z_i$

② number of branches → max(n, m) (pole, zero)

③ symetry (of real axis)

④ RL on the real axis

n+m → laplace

cift → not included (✗)

tek → included (✓)

⑤ Asymtotes

n > m

m > n

angles: $\frac{180}{|n-m|}$

$\frac{180}{4} = 45$ −45 135 −135

intersect:

$$\sigma = \frac{\sum p_i - \sum z_i}{n-m}$$

Auxiliary:

Au. Eq. $A(s)$ $\quad \frac{dA(s)}{ds} = 12s^3 + 18s$

s^5	1	3	−4
s^4	3	9	−12
s^3	0	0	
	12	18	

$$3s^4 + 9s^2 - 12 = 0$$

$$A(s) = 3s^4 + 9s^2 - 12 = 0$$

$s_1 = 1$ $\quad s_3 = +2j$

$s_2 = -1$ $\quad s_4 = -2j$

open loop	steady state error $E(s) = R(s) - Y(s)$	closed loop
$e_{ss} = 1 - G(0)$	$e_{ss} = \lim_{s\to 0} sE(s)$	$\frac{1}{1+G(0)}$

K_p → proportional	error constus	→ $\lim_{s\to 0} G(s)$
K_v → velocity		→ $\lim_{s\to 0} sG(s)$
K_a → acceleration		→ $\lim_{s\to 0} s^2G(s)$

ROOT LOCI

n → zero

n → pole

$1 + G(s)H(s) = 0$ charac. eq.

$G(s)H(s) = -1$ → root locus eq.

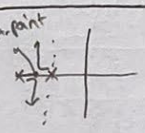
angle eq. $\quad \sum(s-z_i) - \sum(s-p_i)$

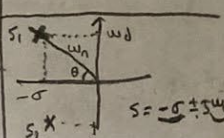
magnitude: $K\frac{\pi|s-z_i|}{\pi|s-p_i|} = 1$

⑥ Breakaway

B. Apoint

$$\sum\frac{1}{\sigma-z_i} = \sum\frac{1}{\sigma-p_i}$$





$\sigma = \zeta\omega_n$ → damping factor

$\omega_d = \omega_n\sqrt{1-\zeta^2}$ (*)

↳ damped frequency

$\theta = \arccos\frac{\sqrt{1-\zeta^2}}{\zeta}$

$0 < \zeta < 1$:

$\zeta > 1$:

$\zeta = 1$:

$\zeta = 0$:

Unit step response

$$y(t) = 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\cdot\sin(\omega_d t + \theta), \quad 0 < \zeta < 1$$

rise time $\quad t_r = \frac{\pi-\theta}{\omega_n\sqrt{1-\zeta^2}}$

peak time $\quad t_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}$

given ω_n, $\zeta \downarrow \to t_r \downarrow$; $\zeta \downarrow \to t_p \downarrow$

given ζ, $\omega_n \uparrow \to t_r \downarrow$; $\omega_n \uparrow \to t_p \downarrow$

overshoot: $\quad M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}}$

$\zeta \uparrow \to M_p \downarrow$

settling time (find) given part

$t_s = \frac{3}{\zeta\omega_n} = \frac{3}{\sigma} = 3\tau = \Delta = \%5$

$t_s = \frac{4}{\zeta\omega_n} = \frac{4}{\sigma} = 4\tau = \Delta = \%2$

$t_s = \frac{5}{\zeta\omega_n} = \frac{5}{\sigma} = 5\tau = \Delta = \%1$ (Error band)

greater K_A:

$\zeta \downarrow$ $\omega_n \uparrow$ $t_p \downarrow$ $M_p \uparrow$

no change t_s

$T(s)$ consider $G(s)$

$$G(s) = \frac{T(s)}{1-T(s)} = \frac{\omega_n^2}{s(s+2\zeta\omega_n)}$$

Adding pole to the open loop

$T_p \uparrow$ $t_r \uparrow$ $M_p \uparrow$

adding pole − closed loop

$T_p \uparrow$ $M_p \downarrow$ $t_r \uparrow$ $S \uparrow$

Adding zero − closed loop

$T_z \uparrow$ $M_p \uparrow$ $t_r \downarrow$ $S \downarrow$

Adding zero − open loop

$T_z \uparrow$ $M_p \uparrow$ ↕ $S_s \uparrow$